Adelaide University

# ARTI5001 Generative Artificial Intelligence

Assignment 3

# Assignment 3 (Group):

# Text-to-Voice (Single-Speaker TTS)

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## Overview

In this group assignment, you will design and implement a **Text-to-Speech (TTS)** system that synthesises **natural and intelligible speech** from input text. You will use the **LJSpeech dataset** as the primary data source and follow an engineering approach that prioritises **reproducibility, fair comparison through controlled experiments, and clear evidence of system behaviour**.

This assignment assesses three components:

- **Code Submission (35%)**: reproducible implementation, controlled experiments, and objective evaluation
- **Report Submission (35%)**: technical communication and evidence-based analysis (paper-style)
- **Presentation & Viva (30%)**: in-person individual contribution and understanding

## Learning objectives

By completing this assignment, your group will demonstrate the ability to:

- build an end-to-end TTS pipeline (**text → waveform**)
- apply suitable dataset handling and pre-processing for speech and text
- design and run **controlled experiments** for fair comparison
- evaluate performance using **objective metric(s)** and qualitative evidence (audio outputs)
- communicate system design, results, limitations, and recommendations in a professional report
- explain and defend technical decisions during an in-person viva

## Required project scope

**Dataset**

- Use **LJSpeech** as the main dataset: https://keithito.com/LJ-Speech-Dataset/
- **Do not upload the dataset** as part of your submission.

**System expectation**

- Produce **playable .wav outputs** from **arbitrary input text**.

- Use a method of your choice (e.g., fine-tuning an existing open-source TTS toolkit/model, or building components yourself), provided the full workflow is clearly implemented and explained.

**Controlled experiments (mandatory)**

- Run **at least two controlled experiments** (e.g., expA and expB):
  - Use the **same dataset split** (same ratio and random seed).
  - Clearly state the **controlled change** between experiments (ideally one major variable).
  - Ensure results are comparable and reproducible.

**Evaluation (mandatory)**

- Report **at least one objective metric** for each experiment (e.g., **latency** or **real-time factor (RTF)**).
- Provide qualitative evidence via **generated audio outputs** (.wav examples).

## Deliverables and marking weights (Total 100 marks)

## 1) Code Submission — 35 marks

Submit one **ZIP file** containing:

1. **One Jupyter Notebook (.ipynb)** containing the full implementation and experiments
2. A folder named **wav_examples/** containing generated .wav outputs

**Recommended ZIP structure (must follow):**

```
<group_number>_A3.zip
├── <group_number>_A3.ipynb
└── wav_examples/
    ├── expA_01.wav
    ├── expA_02.wav
    ├── ...
    ├── expB_01.wav
    ├── expB_02.wav
    └── ...
```

**Minimum requirements inside the notebook**
Include clear evidence for the following:

- Dataset handling and pre-processing (audio + text)
- Train/validation/test split ratio and **random seed** (same split across all experiments)
- At least **two controlled experiments** (expA/expB) with a clearly stated controlled change
- At least **one objective metric** (e.g., latency/RTF) reported for each experiment
- End-to-end inference that synthesises **arbitrary text → playable .wav**, saved into wav_examples/

- Brief environment information (Python version + key packages/versions; CPU/GPU details if applicable)
- In-text citation/acknowledgement where external sources are used (links in comments are acceptable

**Audio examples**

- Include **at least 10 .wav files** generated from **different input sentences** (recommended).
- Clearly label files so the marker can distinguish experiments (e.g., expA_*.wav, expB_*.wav).

## 2) Report Submission — 35 marks

Submit a professional report as <group_number>_A3.pdf, following a paper-style structure (see the provided Report_Template_A3).

**Length**

- Limit the report to **12 A4 pages including references**.
- Prioritise **necessary content and evidence** over strict page count if needed.

**Content requirements (summary)**

- Title page (group details)
- Executive summary (non-technical, ≤ 1 page)
- Introduction and problem framing
- System design (with diagram) and justification
- Experimental design and results table (≥2 controlled experiments)
- Evaluation and analysis (objective metric(s) + qualitative evidence; trade-offs)
- Error analysis and improvement proposals
- Conclusion and recommendations (including scaling considerations)
- References (Harvard)
- Contribution appendix (if required/used by your course practice)

## 3) Presentation & Viva — 30 marks

Submit the presentation slides as <group_number>_A3.pptx. No presentation template is provided.

Each group member must:

- Present a meaningful portion of the project (individual contribution)
- Answer viva questions related to the part presented
- Demonstrate overall understanding of the system and experiments

**Important**

- Presentation/Viva marks are **individual**.
- A member who does not present and/or does not participate in P&V may receive **0 for the Presentation & Viva component**.

**Compute expectations and feasibility**

This project is designed to be achievable on a standard personal laptop **when using appropriate scope choices**, including:

- Training/fine-tuning on a **subset of LJSpeech** (e.g., 1–3k utterances) where required
- Choosing methods that support efficient experimentation and reproducibility

GPU access is **recommended** for faster training and smoother experimentation, but **not strictly required**. If training time is limited, focus on:

- a working end-to-end pipeline,
- fair controlled experiments,
- clear evaluation evidence and analysis.

A full-scale model is not expected. Reasonable scope choices that allow the project to run on a standard laptop are acceptable.

**Academic integrity and use of external resources**

External toolkits, repositories, and tutorials may be used. However:

- Clearly acknowledge external sources where used (links in code comments are acceptable).
- Ensure the notebook demonstrates genuine understanding and meaningful implementation effort.
- The viva will assess individual understanding and contribution.